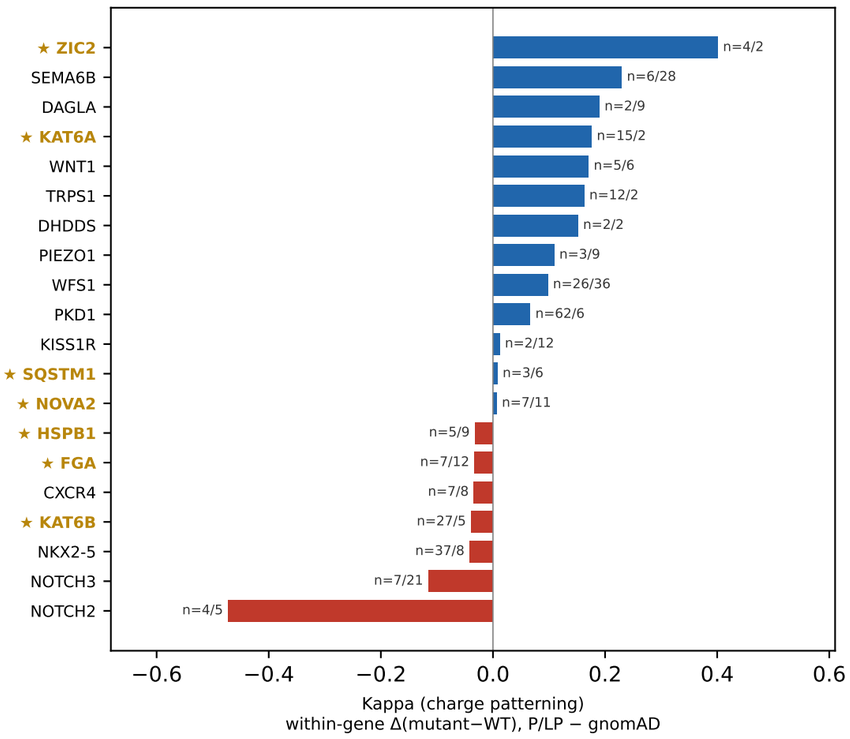
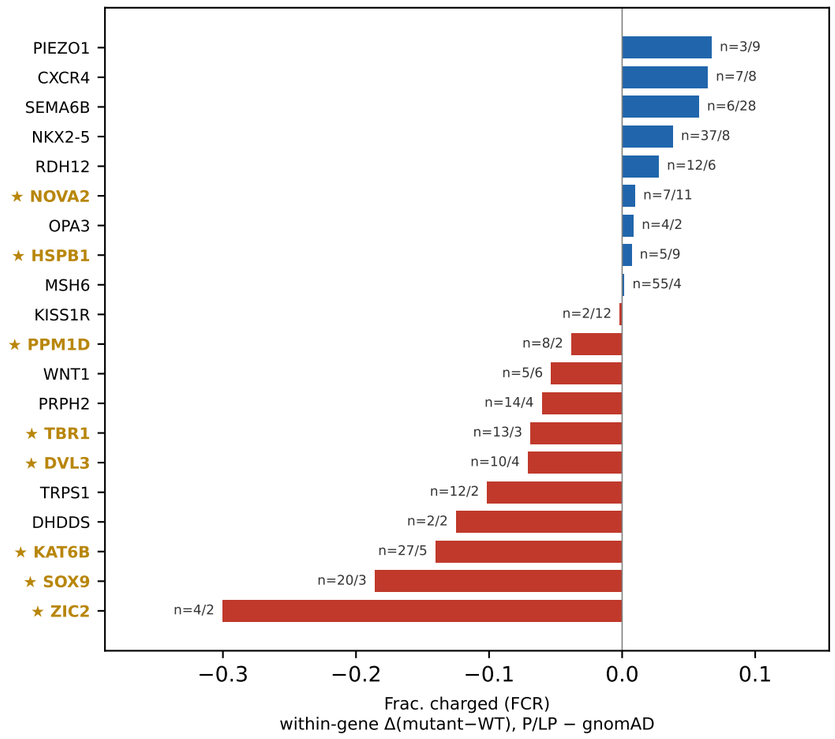
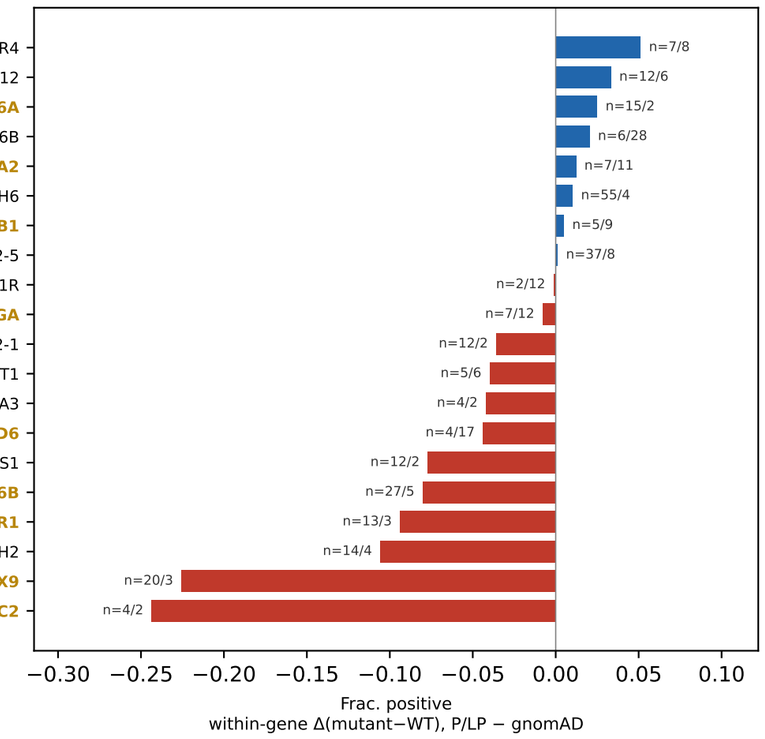
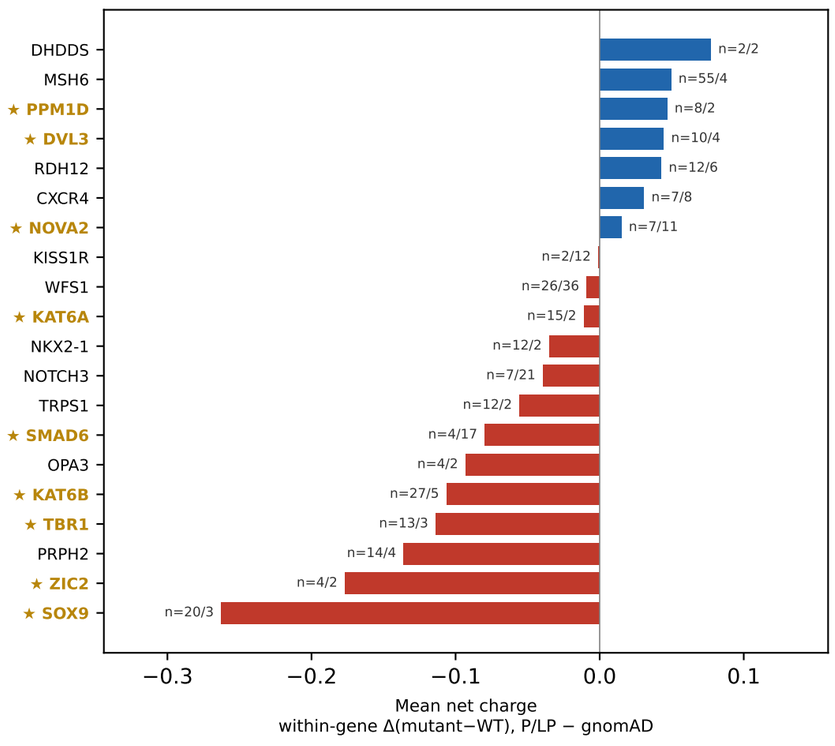
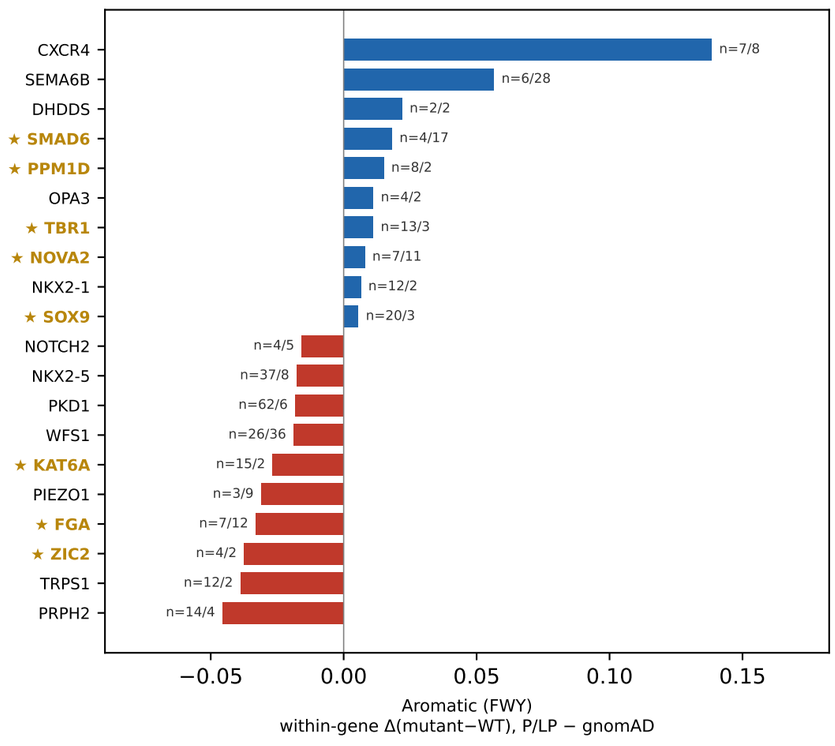
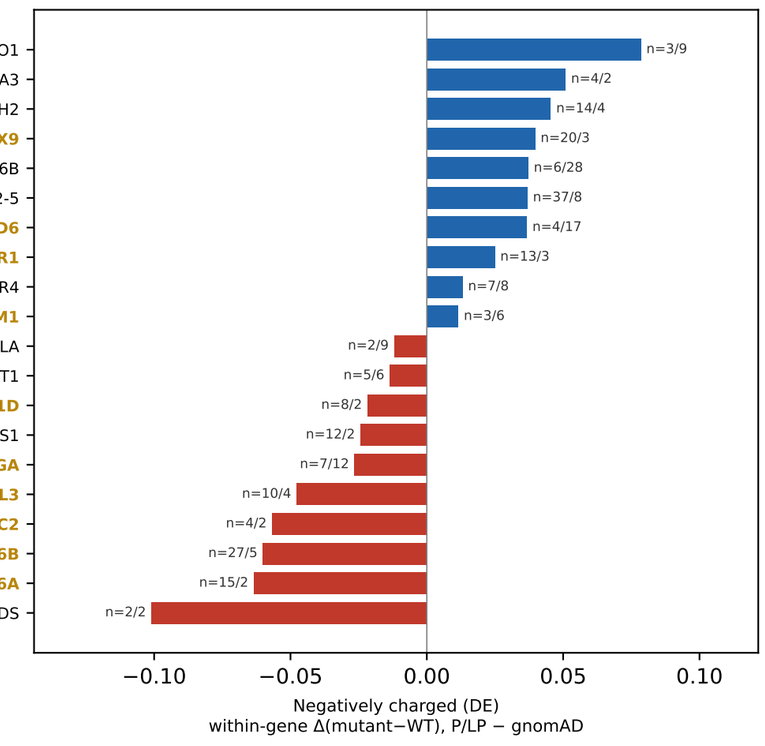

★ ZIC2 n=4/2
SEMA6B n=6/28
DAGLA n=2/9
★ KAT6A n=15/2
WNT1 n=5/6
TRPS1 n=12/2
DHDDS n=2/2
PIEZO1 n=3/9
WFS1 n=26/36
PKD1 n=62/6
KISS1R n=2/12
★ SQSTM1 n=3/6
★ NOVA2 n=7/11
★ HSPB1 n=5/9
★ FGA n=7/12
CXCR4 n=7/8
★ KAT6B n=27/5
NKX2-5 n=37/8
NOTCH3 n=7/21
NOTCH2 n=4/5
−0.6 −0.4 −0.2 0.0 0.2 0.4 0.6
Kappa (charge patterning)
within-gene Δ(mutant−WT), P/LP − gnomAD

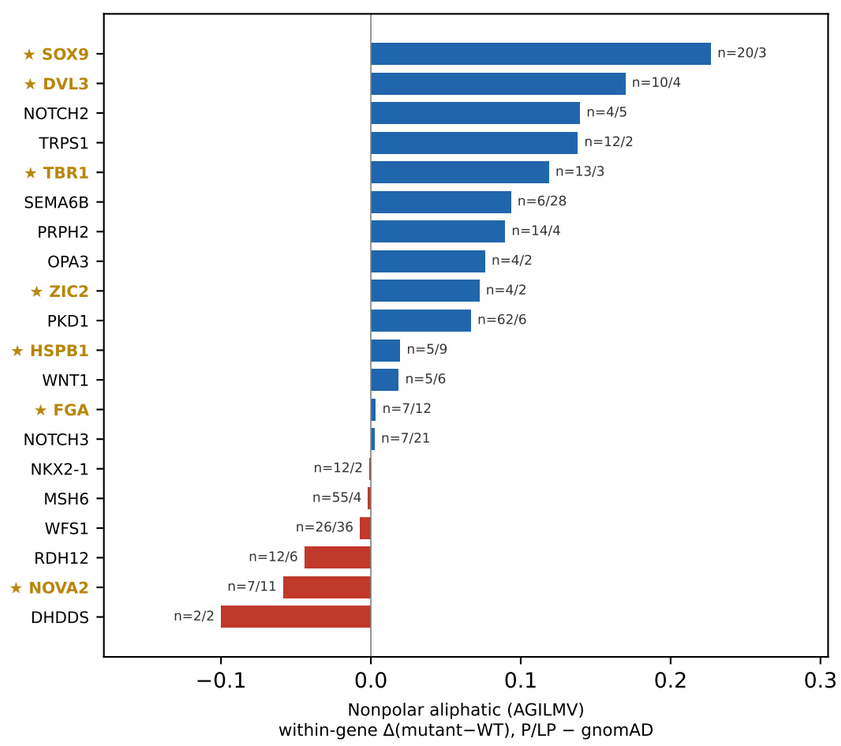

PIEZO1 n=3/9
CXCR4 n=7/8
SEMA6B n=6/28
NKX2-5 n=37/8
RDH12 n=12/6
★ NOVA2 n=7/11
OPA3 n=4/2
★ HSPB1 n=5/9
MSH6 n=55/4
KISS1R n=2/12
★ PPM1D n=8/2
WNT1 n=5/6
PRPH2 n=14/4
★ TBR1 n=13/3
★ DVL3 n=10/4
TRPS1 n=12/2
DHDDS n=2/2
★ KAT6B n=27/5
★ SOX9 n=20/3
★ ZIC2 n=4/2
−0.3 −0.2 −0.1 0.0 0.1
Frac. charged (FCR)
within-gene Δ(mutant−WT), P/LP − gnomAD

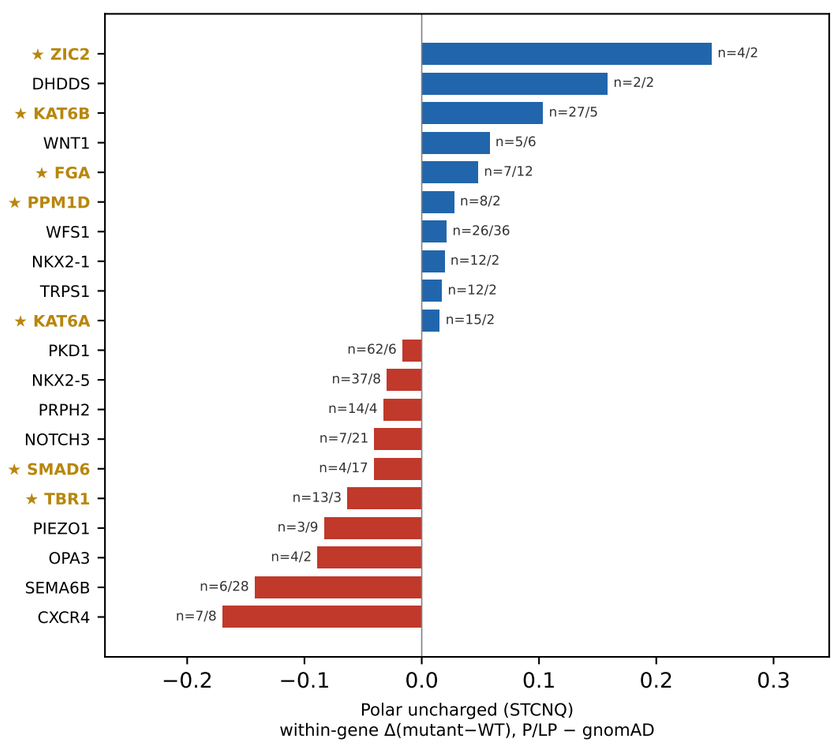

CXCR4 n=7/8
RDH12 n=12/6
★ KAT6A n=15/2
SEMA6B n=6/28
★ NOVA2 n=7/11
MSH6 n=55/4
★ HSPB1 n=5/9
NKX2-5 n=37/8
KISS1R n=2/12
★ FGA n=7/12
NKX2-1 n=12/2
WNT1 n=5/6
OPA3 n=4/2
★ SMAD6 n=4/17
TRPS1 n=12/2
★ KAT6B n=27/5
★ TBR1 n=13/3
PRPH2 n=14/4
★ SOX9 n=20/3
★ ZIC2 n=4/2
−0.30 −0.25 −0.20 −0.15 −0.10 −0.05 0.00 0.05 0.10
Frac. positive
within-gene Δ(mutant−WT), P/LP − gnomAD

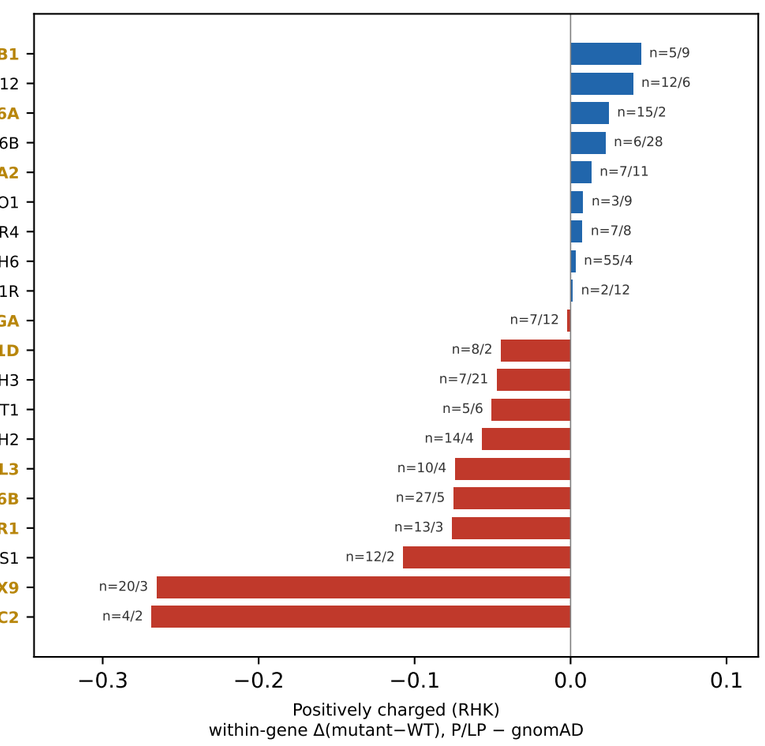

DHDDS n=2/2
MSH6 n=55/4
★ PPM1D n=8/2
★ DVL3 n=10/4
RDH12 n=12/6
CXCR4 n=7/8
★ NOVA2 n=7/11
KISS1R n=2/12
WFS1 n=26/36
★ KAT6A n=15/2
NKX2-1 n=12/2
NOTCH3 n=7/21
TRPS1 n=12/2
★ SMAD6 n=4/17
OPA3 n=4/2
★ KAT6B n=27/5
★ TBR1 n=13/3
PRPH2 n=14/4
★ ZIC2 n=4/2
★ SOX9 n=20/3
−0.3 −0.2 −0.1 0.0 0.1
Mean net charge
within-gene Δ(mutant−WT), P/LP − gnomAD

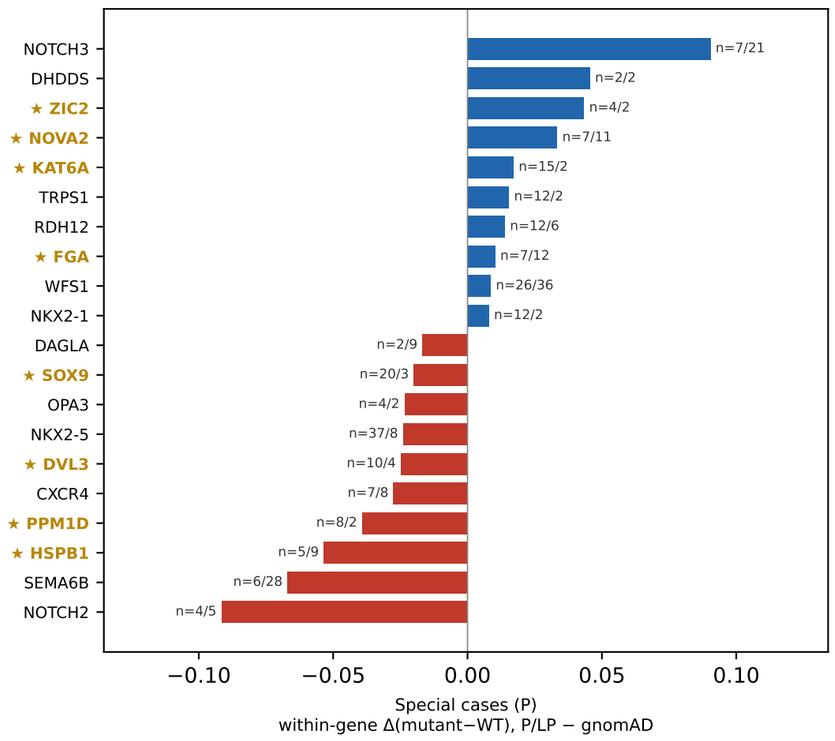

CXCR4 n=7/8
SEMA6B n=6/28
DHDDS n=2/2
★ SMAD6 n=4/17
★ PPM1D n=8/2
OPA3 n=4/2
★ TBR1 n=13/3
★ NOVA2 n=7/11
NKX2-1 n=12/2
★ SOX9 n=20/3
NOTCH2 n=4/5
NKX2-5 n=37/8
PKD1 n=62/6
WFS1 n=26/36
★ KAT6A n=15/2
PIEZO1 n=3/9
★ FGA n=7/12
★ ZIC2 n=4/2
TRPS1 n=12/2
PRPH2 n=14/4
−0.05 0.00 0.05 0.10 0.15
Aromatic (FWY)
within-gene Δ(mutant−WT), P/LP − gnomAD


PIEZO1 n=3/9
OPA3 n=4/2
PRPH2 n=14/4
★ SOX9 n=20/3
SEMA6B n=6/28
NKX2-5 n=37/8
★ SMAD6 n=4/17
★ TBR1 n=13/3
CXCR4 n=7/8
★ SQSTM1 n=3/6
DAGLA n=2/9
WNT1 n=5/6
★ PPM1D n=8/2
TRPS1 n=12/2
★ FGA n=7/12
★ DVL3 n=10/4
★ ZIC2 n=4/2
★ KAT6B n=27/5
★ KAT6A n=15/2
DHDDS n=2/2
−0.10 −0.05 0.00 0.05 0.10
Negatively charged (DE)
within-gene Δ(mutant−WT), P/LP − gnomAD


★ SOX9 n=20/3
★ DVL3 n=10/4
NOTCH2 n=4/5
TRPS1 n=12/2
★ TBR1 n=13/3
SEMA6B n=6/28
PRPH2 n=14/4
OPA3 n=4/2
★ ZIC2 n=4/2
PKD1 n=62/6
★ HSPB1 n=5/9
WNT1 n=5/6
★ FGA n=7/12
NOTCH3 n=7/21
NKX2-1 n=12/2
MSH6 n=55/4
WFS1 n=26/36
RDH12 n=12/6
★ NOVA2 n=7/11
DHDDS n=2/2
−0.1 0.0 0.1 0.2 0.3
Nonpolar aliphatic (AGILMV)
within-gene Δ(mutant−WT), P/LP − gnomAD


★ ZIC2 n=4/2
DHDDS n=2/2
★ KAT6B n=27/5
WNT1 n=5/6
★ FGA n=7/12
★ PPM1D n=8/2
WFS1 n=26/36
NKX2-1 n=12/2
TRPS1 n=12/2
★ KAT6A n=15/2
PKD1 n=62/6
NKX2-5 n=37/8
PRPH2 n=14/4
NOTCH3 n=7/21
★ SMAD6 n=4/17
★ TBR1 n=13/3
PIEZO1 n=3/9
OPA3 n=4/2
SEMA6B n=6/28
CXCR4 n=7/8
−0.2 −0.1 0.0 0.1 0.2 0.3
Polar uncharged (STCNQ)
within-gene Δ(mutant−WT), P/LP − gnomAD


★ HSPB1 n=5/9
RDH12 n=12/6
★ KAT6A n=15/2
SEMA6B n=6/28
★ NOVA2 n=7/11
PIEZO1 n=3/9
CXCR4 n=7/8
MSH6 n=55/4
KISS1R n=2/12
★ FGA n=7/12
★ PPM1D n=8/2
NOTCH3 n=7/21
WNT1 n=5/6
PRPH2 n=14/4
★ DVL3 n=10/4
★ KAT6B n=27/5
★ TBR1 n=13/3
TRPS1 n=12/2
★ SOX9 n=20/3
★ ZIC2 n=4/2
−0.3 −0.2 −0.1 0.0 0.1
Positively charged (RHK)
within-gene Δ(mutant−WT), P/LP − gnomAD


NOTCH3 n=7/21
DHDDS n=2/2
★ ZIC2 n=4/2
★ NOVA2 n=7/11
★ KAT6A n=15/2
TRPS1 n=12/2
RDH12 n=12/6
★ FGA n=7/12
WFS1 n=26/36
NKX2-1 n=12/2
DAGLA n=2/9
★ SOX9 n=20/3
OPA3 n=4/2
NKX2-5 n=37/8
★ DVL3 n=10/4
CXCR4 n=7/8
★ PPM1D n=8/2
★ HSPB1 n=5/9
SEMA6B n=6/28
NOTCH2 n=4/5
−0.10 −0.05 0.00 0.05 0.10
Special cases (P)
within-gene Δ(mutant−WT), P/LP − gnomAD